Arista Access Node Failure SOP (MG-ACC-38, MG Road Core)

Node: MG-ACC-38 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8624

Description

The failure of Arista Access Node MG-ACC-38 at MG Road Core can be attributed to hardware/environmental faults, specifically ASIC memory parity errors, line card crashes, or power supply redundancy loss. This is due to the inherent design of the Arista 7500R series switch, which utilizes a distributed architecture and relies on the ASIC's memory to store routing information and forwarding tables. When an ASIC memory parity error occurs, the node may crash, leading to a critical failure. Similarly, a line card crash can occur due to overheating, electrical noise, or software faults, while power supply redundancy loss can be caused by a faulty power source or a malfunctioning power supply unit (PSU). The ALARM_8624 alarm code is triggered in response to these failures, indicating a critical hardware or environmental issue that requires immediate attention.

Pre-requisite Checks

- Confirm that the MG Road Core site is accessible and that remote hands are dispatched to the location.

- Verify the status of the redundant links and ensure that they are functioning properly.

- Check the power supply units (PSUs) for any signs of malfunction or failure.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System Resource Information, Description: This command displays information about the system's resources, including memory usage, CPU utilization, and disk space. It helps identify potential issues with the ASIC's memory or line cards.

Command: show board detail, Expected Output: Board Detail Information, Description: This command provides detailed information about the line cards, including their status, temperature, and power consumption. It helps identify potential issues with the line cards or power supplies.

Command: show power supply, Expected Output: Power Supply Information, Description: This command displays information about the power
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supplies, including their status, output voltage, and current
consumption. It helps identify potential issues with the power
supplies or PSUs.
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Resolution Steps

Step 1: Step: Replace faulty ASIC, Command: replace system resource , Description: This command replaces the faulty ASIC with a new one, resolving the memory parity error issue.

Step 2: Step: Replace faulty line card, Command: replace board , Description: This command replaces the faulty line card with a new one, resolving the line card crash issue.

Step 3: Step: Replace faulty power supply, Command: replace power supply , Description: This command replaces the faulty power supply with a new one, resolving the power supply redundancy loss issue.

Post-Check Verification

```
Command: show system resource, Expected Output: System Resource
Information, Description: This command verifies that the system
resources are functioning properly after the resolution.

Command: show board detail, Expected Output: Board Detail Information,
Description: This command verifies that the line cards are functioning
properly after the resolution.

Command: show power supply, Expected Output: Power Supply Information,
Description: This command verifies that the power supplies are
functioning properly after the resolution.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy running-config startup-config, Description: This command restores the original configuration, reverting any changes made during the resolution process.

- Step: Replace original hardware, Command: replace , Description: This command replaces the original hardware with the original hardware, reverting any changes made during the resolution process.